Worksheet — 1.1 Processors, Input, Output & Storage

OCR A Level Computer Science (H446) — Component 01, Section 1.1 (1.1.1 processor, 1.1.2 processor types, 1.1.3 I/O & storage).

Name: __ **Date:** ____

Time: ~60 min. Write your answers in the spaces; check against the answer key at the end.

Section A — Skills practice

1. Key-term match

Match each term (1–8) to its definition (A–H). Write the letter on the line.

Terms

1. Register .... ____
2. Cache .... ____
3. Pipelining .... ____
4. Core .... ____
5. Volatile .... ____
6. Virtual memory .... ____
7. GPU .... ____
8. Control Unit .... ____

Definitions

- **A.** Overlapping the fetch, decode and execute of consecutive instructions to raise throughput.
- **B.** Loses its contents when power is removed.
- **C.** Small, very fast store inside the CPU that holds data during processing.
- **D.** Many-core processor that performs the same operation on large data sets in parallel.
- **E.** Use of secondary storage as extra RAM when physical RAM is full.
- **F.** An independent processing unit able to run its own fetch–decode–execute cycle.
- **G.** Small, fast memory on or near the CPU holding frequently used data/instructions.
- **H.** Decodes instructions, generates control signals and synchronises the CPU with the clock.

2. Fill in the blanks — the Fetch–Decode–Execute cycle

Complete the passage using the word bank. Each word is used once.

Word bank: opcode · operand · PC · MDR · CIR · data · control · increment · ALU · address

During **fetch**, the address of the next instruction is copied from the __ **into the MAR. A read signal is sent on the _** ***bus, and the instruction returns along the _*** **bus into the __. The processor then _*****s the PC so it points to the next instruction, and the fetched instruction is copied from the MDR into the _*****.**

During **decode**, the Control Unit splits the instruction into the __ **(the operation to perform) and the _** ***(the data or*** ___ to act on).

During **execute**, the instruction is carried out — for example a calculation is performed by the ______ with the result stored in the accumulator.

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3. State the register

For each description, name the register (PC, ACC, MAR, MDR or CIR).

a) Holds the address of the **next** instruction to be fetched. .... ______

b) Holds the **address** of the location currently being read from or written to. .... ______

c) Holds the **data or instruction** just fetched from (or about to be written to) memory. .... ______

d) Holds the **current instruction** while it is being decoded and executed. .... ______

e) Stores the **immediate result** of an ALU calculation. .... ______

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4. Put the FDE steps in order

Number these fetch-stage steps 1–4 in the order they happen.

____ Read signal placed on the control bus; instruction returns via the data bus into the MDR.

____ The contents of the MDR are copied into the CIR.

____ The address in the PC is copied into the MAR.

____ The PC is incremented (PC := PC + 1).

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5. Short answer — performance & architectures

a) State **two** factors that affect CPU performance and explain the effect of each.

b) "Adding more cores always makes a computer faster." Give the **caveat** that makes this statement only partly true.

c) Explain the key difference between **Von Neumann** and **Harvard** architectures, and give one typical use of each.

d) Give **two** differences between **CISC** and **RISC** processors.

e) Explain why a GPU is well suited to graphics and machine-learning work but poor at branch-heavy sequential tasks.

6. Storage device selection table

Complete the empty cells. For each storage type, state **how it works**, **one pro**, **one con** and **one example use**.

Type	How it works	One pro	One con	Example use
Magnetic (e.g. HDD, tape)				
Flash/SSD (e.g. SSD, USB, SD)				
Optical (e.g. CD, DVD, Blu-ray)				

7. Challenge box

Challenge — virtual memory & thrashing. A user keeps many large applications open and the computer becomes very slow, with the hard disk constantly active.

(i) Explain what **virtual memory** is and why the OS is using it here.

(ii) Name the problem causing the slowdown and explain **why** moving pages between RAM and disk is so slow.

(iii) Suggest **one** hardware change that would reduce the problem, and justify it.

8. Scenario table — pick the right processor/architecture/storage

Examiners consistently report that candidates can *define* architectures and storage types but cannot *select* the right one for a scenario. For each scenario below, choose the best option from the bank and give a **one-line justification tied to the scenario** (a generic definition scores nothing).

Bank: Von Neumann multicore CPU · Harvard-architecture microcontroller · RISC processor · GPU · Magnetic tape · Flash storage (SSD)

Scenario	Choice	Justification (linked to the scenario)
a) A washing machine's controller runs one small, fixed program that never changes, kept separate from its working data.		
b) A data-science team trains machine-learning models that multiply enormous matrices of numbers.		
c) A battery-powered fitness watch must run for a week between charges.		
d) A national archive must keep 800 TB of records off-site as cheaply as possible; they are almost never read back.		
e) A video journalist needs portable storage that survives being dropped and transfers footage quickly in the field.		
f) An office desktop must run many different applications, loading new programs and data into the same memory.		

9. FDE register transfers — order the notation

The register-transfer steps of one full fetch–decode–execute cycle are shown out of order. Number them **1–6**. (`memory[MAR]` means "the contents of the memory location whose address is in the MAR".)

_____ CIR ← MDR

_____ Decode: the CU splits the contents of the CIR into opcode and operand

_____ MAR ← PC

_____ Execute: the ALU carries out the operation; any result is stored in the ACC

_____ MDR ← memory[MAR]

_____ PC ← PC + 1

Section B — Exam practice (30 marks)

Answer all questions in full sentences where appropriate. Marks and assessment objectives are shown in brackets.

Q1 State the purpose of the Memory Data Register (MDR). **[1]** (AO1)

Q2 Describe how pipelining improves the performance of a processor. **[2]** (AO1)

Q3 A supermarket is installing self-service checkouts. Identify **three** input or output devices the checkout will need, and state the purpose of each **in this scenario**. **[3]** (AO2)

Q4 A wildlife photographer works on location, carries her storage in a rucksack, and transfers thousands of large image files to her drive every day. Explain **two** reasons why flash storage (an SSD) is more suitable than a magnetic hard disk drive for this photographer. **[4]** (AO2)

Q5 The Program Counter of a processor holds the address 0064. Memory location 0064 contains the instruction `ADD 0100`. Complete the table to show the contents of each register **once the fetch stage of the cycle is complete**. **[4]** (AO2)

Register	Contents after fetch
PC	
MAR	
MDR	
CIR	

Q6 Explain how increasing the amount of cache can improve the performance of a CPU. **[3]** (AO1)

Q7 Describe **two** differences between RISC and CISC processors. **[4]** (AO1)

Q8* An animation studio is buying new workstations. The workstations will spend most of their time rendering complex 3D scenes, but staff will also use them for everyday office tasks such as email and word processing. The studio must choose how to spend a fixed budget across: a higher clock speed, more CPU cores, more cache, and a dedicated GPU.

Discuss the factors the studio should consider when specifying the processors for these workstations, and recommend how the budget should be spent. **[9]** (AO3)

The quality of your extended response will be assessed in this question.